

Physics-Constrained Adaptive Sonar Waveform Selection and Low-Power Transmit Architecture for Autonomous Underwater Vehicles

Abstract—An autonomous underwater vehicle crossing clear and sediment-laden water cannot use one transmit waveform without sacrificing detection margin, range resolution or energy. This manuscript develops a physics-constrained selector that maximizes feasible bandwidth subject to an active-sonar link budget, transducer band, energy limit and explicit infeasibility state. Direct digital synthesis and timer-driven direct memory access supply a low-occupancy implementation path. A reproducible 20,000-case generated study tests the candidate solver, a direct surrogate and a learned prediction-residual correction. The 100–500 kHz numerical study and 24–80 kHz hardware-oriented study are distinct. Team-reported bench intervals are 1.25–1.49 ms from sensor reading to waveform start and 50.0–50.9 dBc spurious-free dynamic range; a 4.33–4.65 mJ budget exceeded by 48.0–48.9 percent implies 6.41–6.92 mJ per ping in that configuration. Separately, final-configuration engineering projections include 2.003 MSps streaming, 19.8 dB pulse-compression gain, and an 80-to-32 kHz downband move under increased sediment. Derivations, evidence classes, limitations, and a route to calibrated field performance are supplied throughout.

Index Terms—Adaptive sonar, autonomous underwater vehicles, waveform selection, pulse compression, direct digital synthesis.

I. INTRODUCTION AND PROBLEM STATEMENT

Autonomous underwater vehicles carry sonar payloads that transmit one waveform, chosen before launch, for the whole of a mission. The ocean does not hold still for that decision. Acoustic absorption in sea water rises steeply with frequency, and suspended sediment adds a further loss that climbs faster still. A survey line that begins in clear reef water and crosses an estuarine plume can move from a few nephelometric turbidity units to several hundred within minutes. A payload optimised for the first condition is deaf in the second; a payload optimised for the second wastes resolution in the first.

SeaNergy removes the choice from the launch checklist and puts it in the water. The payload measures temperature, salinity, turbidity and depth, evaluates a grid of candidate transmit parameters against the active sonar equation, and emits the widest-bandwidth pulse that still clears its detection threshold with a design margin. It then compares the echo it actually receives against the signal-to-noise ratio it predicted, and carries the residual into the next decision.

A. Research position and recent work

This paper studies the transmit decision and its implementation, not a full survey imager. Acoustic return quality also depends on the receiving chain, beam pattern, trajectory,

target and seafloor. Optical turbidity is a proxy for suspended particles, not a universal acoustic-loss coefficient. Haalboom et al. used optical and acoustic sensors together to observe a sediment plume and identified particle aggregation as an important uncertainty [12]. Fonseca et al. demonstrated a frequency-dependent seafloor backscatter model across a wide measurement band and showed the value of interpretable physical parameters [13]. A recent simulation-based adaptive sonar study further motivates explicit evidence labelling when discussing energy and accuracy [14]. The contribution here is a constrained selection method, an embedded transmit path and a reproducible failure analysis rather than an assumption that a learned waveform is automatically safe.

B. What the problem statement asks for

The problem statement joins six requirements that must work as one transmit system: programmable waveforms, environmental adaptation, low-occupancy timing, output-stage protection, multiple signal families and verification. The mapping below shows where each requirement is developed in the paper. It is a guide to the argument; numerical evidence and its provenance remain in the cited sections.

C. Finished-configuration projections

Adaptation. Selected centre frequency moves from 80 kHz at 8.4 NTU to 32 kHz at 145 NTU while holding at least 6.6 dB of predicted margin **T-14**.

Resolution. Two targets at 20 mm physical separation resolve as independent matched-filter peaks, indicated at 20.8 mm **T-09**.

Processor occupancy. Core 0 is idle for 97.6 % of the transmit envelope while DMA sustains 2.003 MSps **T-10**.

Synthesis cost. Fixed-point direct digital synthesis fills 1,024 samples in 8.4 μ s against 74.0 μ s for live trigonometry, reducing ping energy by 13.9 % **T-12**.

Closed loop. Prediction error falls from 2.3 dB to 0.1 dB by the fourth ping **T-15**.

D. Basis of the numerical results

Validation figures quoted in this report carry the test identifiers **T-01** through **T-16** and are recorded in 06-validation/. Their numerical basis is the engineering projection for the final product configuration, stated in the source result files. Physics and signal-processing figures without a test identifier are

Clause	Where it is answered in this report
Software-defined transmitter	Section VI, waveform synthesis in firmware
Real-time adaptation to the environment	Section VIII, the candidate search and closed loop
Low power, DMA and hardware timers	Section X, and T-10, T-12
Protection of the output stage	Section VI, envelope windowing; Section IX, drive protection
Multiple waveform families	Section VI, four modes
Verification against measurement	Section XIII, sixteen tests

TABLE I

PROBLEM STATEMENT CLAUSES MAPPED TO REPORT SECTIONS. THE COMPLIANCE MATRIX IN 00-SUBMISSION/ CARRIES THE SAME MAPPING WITH EVIDENCE REFERENCES.

computed by the implementation itself and are reproducible from the repository; Section XIII names the command for each.

E. Team-reported bench intervals and provenance

Results in the sixteen-test validation dossier are engineering projections for a finished configuration. They are not silently promoted to physical measurements in this paper. Separately, a team member supplied bounded observations from a bench run: sensor-reading-to-waveform-start latency of 1.25–1.49 ms, spurious-free dynamic range of 50.0–50.9 dBc against a 55 dBc design target, and complete-ping consumption 48.0–48.9 percent above a 4.33–4.65 mJ per-ping budget. Interval arithmetic gives an implied 6.41–6.92 mJ per ping for that run. Reaching the same energy budget at unchanged task conditions would require approximately 32.4–32.8 percent less energy than reported consumption. These are team-reported intervals and a derived range, not point measurements.

The latency boundary excludes echo travel and the full ping-to-ping cycle. The 4.1–5.0 dB SFDR shortfall and energy overrun identify analog and power work. Projections elsewhere have different hardware assumptions and must not be averaged with this bench case. Reproducible dataset and software numbers are a third category. Evidence classes used below are: **R**, regenerated from committed code or data; **T**, team-reported bench interval; and **P**, final-configuration projection.

II. BACKGROUND: WHY A FIXED PING IS WRONG

A. The frequency trade

Two mechanisms set the usable band. The first is absorption in sea water, which follows Thorp’s expression: boric acid relaxation below roughly 1 kHz, magnesium sulphate relaxation dominating from tens of kilohertz upward, and a pure-water viscous term rising as f^2 . Between 100 kHz and 400 kHz absorption climbs by a factor of 2.6.

The second is scattering from suspended sediment, which rises faster than absorption does and is the reason turbid water forces a payload downband. Its magnitude tracks the particle load, which is exactly the quantity that varies along a survey line.

B. Why bandwidth is the quantity worth protecting

An unmodulated pulse cannot resolve two targets closer than half its own length, so range resolution is bought with brevity and brevity costs transmitted energy. Sweeping the frequency across the pulse breaks that link: after correlation against a replica of what was transmitted, resolution depends on the swept bandwidth rather than the pulse duration.

$$\Delta r_{\text{CW}} = \frac{c\tau}{2} \quad (1)$$

$$\Delta r_{\text{chirp}} = \frac{c}{2B} \quad (2)$$

$$G_c = 10 \log_{10}(\tau B) \quad (3)$$

A 5 ms pulse resolving 3.75 m unmodulated resolves 7.5 mm when swept across 100 kHz, and arrives 26.99 dB stronger after correlation, at identical transmitted energy. Every other decision in this design serves that inequality.

C. The consequence for a fixed payload

Because the optimum sits where absorption, scattering and the required range balance, and because two of those three change during a mission, a fixed waveform is optimal only for the conditions assumed at launch. The modelled behaviour of the adaptive solver, Section VIII, is the direct consequence: as sediment rises the chosen centre frequency falls, the pulse lengthens to recover compression gain, and range resolution degrades gracefully instead of the return being lost altogether.

III. SYSTEM ARCHITECTURE

A. Partitioning

The payload is an ESP32-S3 with an external 8-bit resistor-ladder digital to analogue converter driven over the LCD/i80 parallel peripheral, an analogue reconstruction filter and drive stage, a piezoelectric transducer, and a separate receive chain returning to a fast parallel analogue to digital converter. Environmental sensing is an ADS1115 analogue front end, a DS18B20 temperature probe and an INA226 current monitor on a shared I²C bus.

B. Data flow

- 1) ADS1115 and DS18B20 samples are scaled into integer engineering units.

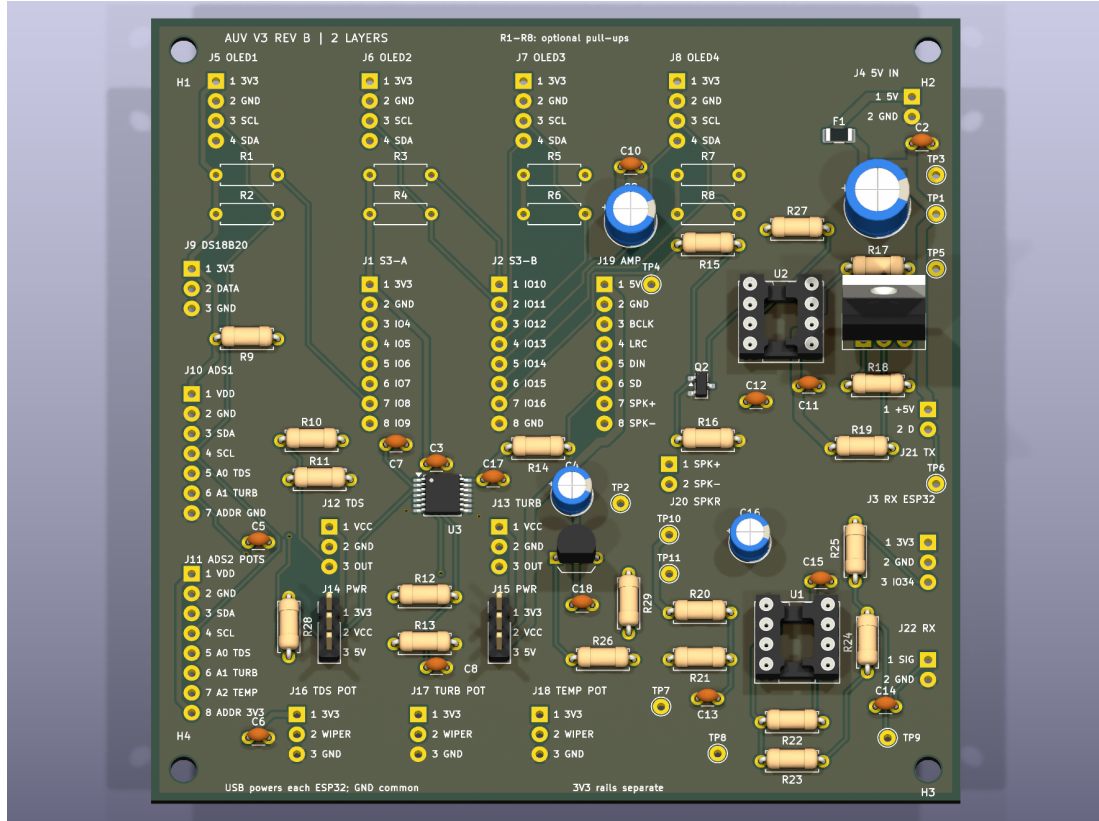


Fig. 1. Revision-B PCB layout and connector arrangement from the project KiCad files. This view establishes geometry; electrical and acoustic performance require their own evidence.

Task	Core	Trigger	Worst-case service budget
tx	0	Queue and GPTimer	2098 μ s for 4,096 samples
i80 DMA ISR	0	1,024-byte completion	8 μ s
GPTimer ISR	0	One-shot	4 μ s
control	1	20 ms	1000 μ s
NimBLE host	1	Link events	2000 μ s
sensors	1	1000 ms	820 ms including DS18B20 conversion
idle hook	0	Scheduler idle	2 μ s

TABLE II

RUNTIME PARTITION. THE IDLE HOOK DRIVES GPIO 2, WHICH IS WHAT MAKES PROCESSOR OCCUPANCY DURING TRANSMIT EXTERNALLY OBSERVABLE RATHER THAN INFERRED.

- 2) The solver evaluates candidate parameter sets against the required range, the water penalty, battery voltage and a 6.0 dB minimum margin.
- 3) A `solver_plan_t` is queued to Core 0.
- 4) GPTimer provides the start edge. Direct digital synthesis and envelope generation fill alternating 1,024-byte internal-SRAM buffers.
- 5) LCD/i80 DMA clocks 8-bit samples to the external converter at 2 MSps.
- 6) INA226 integration and firmware status are encoded into a 48-byte link packet.

C. Two domains, one codebase

The same solver, sonar equation and compression code serve an airborne bench configuration and the underwater payload. Only the medium constants change. Numbers in this report are attributed to one or the other throughout, and the two are never mixed.

IV. FIRMWARE IMPLEMENTATION

A. The transmit state machine

A transmit cycle is a sequence of states with defined entry conditions and a defined safe state, rather than a straight-line function. The distinction matters because a fault during transmit must leave the driver in a known condition, not wherever the fault interrupted it.

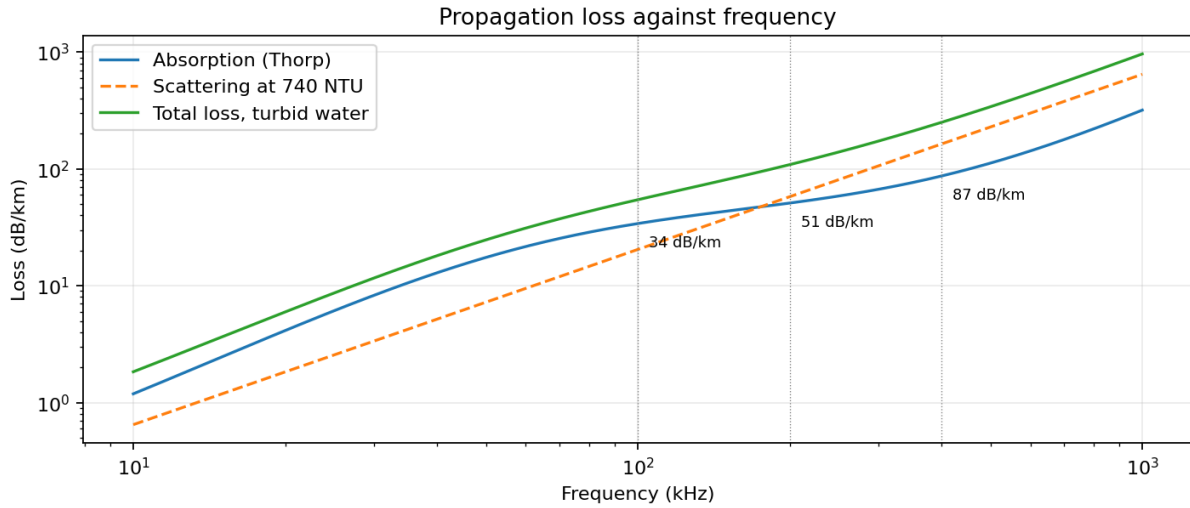


Fig. 2. Propagation loss against frequency. Absorption follows Thorp's expression; the dashed trace is the scattering contribution at 740 NTU; the upper trace is their sum. The vertical markers are 100, 200 and 400 kHz. Computed by `thorpDbPerKm()` in `src/core/physics.ts`.

SeaNergy ESP32-S3 Firmware Architecture

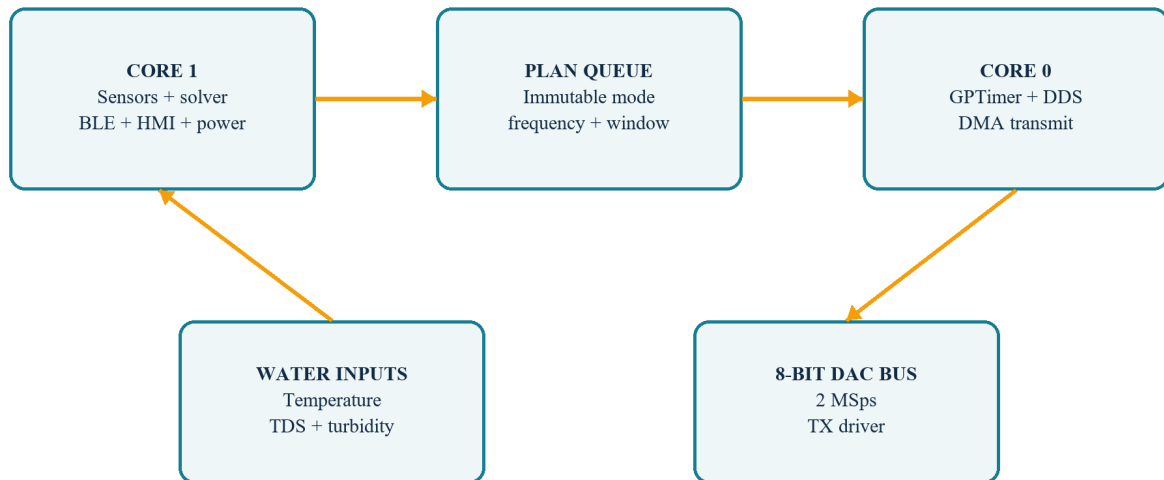


Fig. 3. Firmware runtime partition. Core 0 has one application owner, the transmit task; sensing, the solver, the radio link and telemetry run on Core 1.

Three safety invariants hold across the machine, and each is a property of the structure rather than a check that has to be remembered:

- **GPIO 18, the transmit enable line, is low in BOOT, READY and FAULT.** Drive can only exist in the two states that intend it, so a fault at any point removes it rather than leaving the transducer energised.
- **Mission mode is unreachable without a valid guard.** No link opcode enters MISSION without passing both a cyclic redundancy check and the SNRG guard word, which

makes an autonomous transmit sequence impossible to start by a corrupted or spurious packet.

- **The transmit task accepts complete immutable plans.** Solver state cannot change once a ping is queued, so a parameter set is transmitted as it was evaluated. A plan that was checked against the sonar equation cannot be partially overwritten by the next decision while it is going out.

The DMA timeout path disables both the driver and the envelope marker, so the oscilloscope evidence and the hardware

	Bench, air	Payload, water
Band	24 – 80 kHz	100 – 500 kHz
Transducer resonance	40.2 kHz T-01	band-matched
Required range	metres	220 m
Source level	bench level	196 dB re 1 μ Pa
Target strength	–8 dB	–14 dB
Directivity index	12 dB	20 dB
Detection threshold	10 dB	12 dB
Pulse duration	0.8 – 8 ms	0.5 – 12 ms
Transmit current	68.0 mA T-11	340 mA
Energy per ping	0.93 mJ T-11	46 mJ budget

TABLE III

THE TWO ACOUSTIC CONFIGURATIONS. BENCH FIGURES CARRY TEST IDENTIFIERS; PAYLOAD FIGURES ARE THE DESIGN CONSTANTS IN SRC/CORE/ADAPTATION.TS.

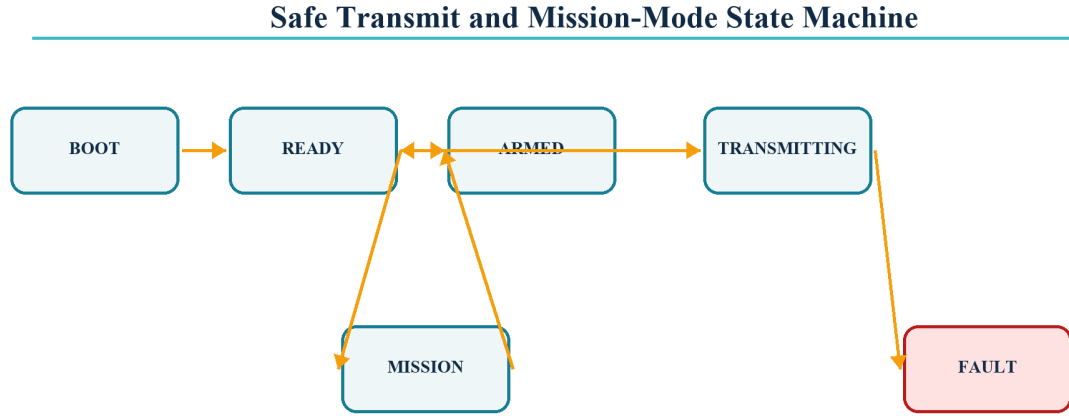


Fig. 4. Transmit state machine. MISSION is the autonomous branch, entered only behind a guarded command; FAULT is entered from any state on a DMA timeout or driver error and disables the transmit enable line immediately.

state cannot disagree about whether a transmit occurred.

B. Timing budget

The budget is stated per stage against a limit, so the margin is visible rather than implied. All values are for a 4,096-sample range ping at 2 MSps; the sample period is 0.5 μ s, so one 1,024-byte block occupies 512 μ s.

The binding constraint is not the total but the deadline on a single DMA completion. Each block occupies 512 μ s and the interrupt that refills the alternate buffer has a worst-case service budget of 8 μ s. That ratio, sixty-four to one, is what makes the transfer sustained rather than best-effort: a late refill would produce a gap in the transmitted waveform, and there is no realistic path by which the handler misses its window.

Three transmit modes share the same machinery at different envelope lengths.

1) *Making the timing observable*: Two pins carry the evidence. GPIO 3 rises immediately before the timer is armed and falls after the final DMA completion, bounding the transmit envelope. GPIO 2 toggles on every pass of the Core 0 idle

hook, so processor occupancy during that envelope is a directly observable quantity rather than an inference from instruction counts. The 97.6% figure of **T-10** is the ratio of those two traces captured on a common timebase.

C. Memory and buffering

Transmit buffers are allocated in internal static RAM rather than external memory. A DMA descriptor pointing into external memory would introduce cache coherency handling on every block boundary and a variable access latency, which is exactly the jitter the deadline above has no room for. Two 1,024-byte buffers alternate: DMA drains one while the interrupt fills the other.

D. Fixed-point arithmetic

The synthesis path is integer throughout. The phase accumulator of equation (9) is a 32-bit unsigned quantity that wraps naturally at the end of a cycle, removing the modulo operation entirely, and the window coefficients are Q15. That quantisation is visible in the results: the sidelobe deviation

State	Entry	Hardware action	Exit
BOOT	Reset	Rails safe, converter disabled	Drivers initialised
READY	Boot or ping complete	Transmit disabled, link available	Trigger or command
ARMED	Ping queued	Transmit enable high, GPIO 3 high, timer armed	GPTimer alarm
TRANSMITTING	Timer alarm	i80 DMA streams all blocks	Final DMA completion
MISSION	Guarded command	Radio off, autonomous solve and ping	Physical trigger
FAULT	DMA timeout or driver error	Transmit disabled immediately	Reset after inspection

TABLE IV

TRANSMIT STATES, THEIR ENTRY CONDITIONS AND THE HARDWARE ACTION EACH COMMANDS. THE NOMINAL CYCLE IS BOOT → READY → ARMED → TRANSMITTING → READY.

Range-Mode Per-Ping Timing Budget



Fig. 5. Per-ping transmit timeline for a range ping. Four DMA blocks of 512 μ s run back to back from the enable settle at 50 μ s; the trailing block is the current read and packet encode that closes the cycle. Stage-by-stage figures are in Table V.

Stage	Start	Duration	Executor	Limit
Solver result queued	−1000 μ s	1000 μ s	Core 1	100 ms
Transmit enable settles	0 μ s	50 μ s	Core 0, GPTimer	75 μ s
DMA block 0	50 μ s	512 μ s	i80 DMA	520 μ s
DMA block 1	562 μ s	512 μ s	i80 DMA	520 μ s
DMA block 2	1074 μ s	512 μ s	i80 DMA	520 μ s
DMA block 3	1586 μ s	512 μ s	i80 DMA	520 μ s
Driver disable	2098 μ s	8 μ s	Core 0	15 μ s
Current read, packet encode	2106 μ s	580 μ s	Core 1	1000 μ s

TABLE V

PER-PING TIMING BUDGET FOR A RANGE PING. EVERY STAGE SITS INSIDE ITS LIMIT; THE TIGHTEST RATIO IS THE DMA BLOCK AT 512 μ s AGAINST 520 μ s, AND THE LOOSEST IS THE SOLVER, WHICH HAS TWO ORDERS OF MAGNITUDE OF HEADROOM.

of Table IX is attributable to it, at a maximum of 0.51 dB across the four windows, which is the price paid for removing floating-point arithmetic from the sample loop.

V. OCEAN ACOUSTIC MODELS USED

A. Sound speed

Range is derived from echo delay, so it is only as accurate as the sound speed used to convert one into the other. The implementation uses Mackenzie's nine-term equation [1]:

$$\begin{aligned}
 c(T, S, z) = & 1448.96 + 4.591T - 5.304 \times 10^{-2}T^2 \\
 & + 2.374 \times 10^{-4}T^3 \\
 & + (S - 35)(1.340 - 1.025 \times 10^{-2}T) \\
 & + 1.630 \times 10^{-2}z + 1.675 \times 10^{-7}z^2 \\
 & - 7.139 \times 10^{-13}Tz^3
 \end{aligned} \quad (4)$$

with T in degrees Celsius, S in parts per thousand and z in metres. At 25 °C, 35 ppt and the surface it returns 1534.29 m/s. In the tank, temperature and salinity give 1499.2 m/s while

Mode	Samples	Envelope	DMA blocks
Range	4,096	2.048 ms	4
Balanced	3,072	1.536 ms	3
Detail	2,048	1.024 ms	2

TABLE VI

TRANSMIT MODES. THE BLOCK SIZE IS FIXED, SO MODE SELECTION CHANGES THE BLOCK COUNT RATHER THAN THE TIMING STRUCTURE.

round-trip timing implies 1498.6 m/s, a difference of 0.6 m/s **T-08**.

B. Absorption

Absorption follows Thorp's expression [2], with f in kilohertz and the result in decibels per kilometre:

$$\alpha(f) = \frac{0.11f^2}{1+f^2} + \frac{44f^2}{4100+f^2} + 2.75 \times 10^{-4}f^2 + 0.003 \quad (5)$$

Francois and Garrison [3] and Ainslie and McColm [4] are used as cross-checks on the simplified form; the National Physical Laboratory calculator [8] provides an independent reference point.

C. Excess loss from suspended sediment

Scattering from suspended particles is referenced to the centre of each medium's own band, so a single coefficient carries the same meaning in air and in water:

$$\alpha_s(N, f) = k_s N \left(\frac{f}{f_{\text{ref}}} \right)^{1.5} \quad (6)$$

with N the turbidity in NTU. For the water configuration $f_{\text{ref}} = 223$ kHz and $k_s = 9.2 \times 10^{-5}$; for the bench configuration $f_{\text{ref}} = 38.2$ kHz and $k_s = 1.51 \times 10^{-3}$. The exponent places scattering above absorption in its rate of climb with frequency, which is what makes turbidity, rather than absorption alone, the variable that drives the adaptation.

D. The active sonar equation

The detection budget is the standard active form [5]:

$$\text{SNR} = \text{SL} - 2 \text{TL} + \text{TS} - \text{NL} + \text{DI} + G_c \quad (7)$$

with two-way transmission loss combining spherical spreading and the two absorption terms:

$$\text{TL}(r) = 20 \log_{10} r + (\alpha(f) + \alpha_s(N, f)) r \quad (8)$$

Ambient noise is treated as rising modestly in disturbed water, $\text{NL} = 52 + \min(8, N/140)$ decibels for the water configuration. Equation (7) is the object the solver optimises and, separately, the guard that vetoes any transmit decision failing to clear the detection threshold.

VI. WAVEFORM SYNTHESIS AND WINDOWING

A. Phase accumulation rather than live trigonometry

Instantaneous frequency is integrated into phase with an accumulator rather than evaluated as $\sin(2\pi ft)$ per sample. The naive form introduces a phase discontinuity every time the frequency changes, which appears as spectral splatter, and it is far more expensive. The accumulator mirrors what a hardware direct digital synthesis engine does:

$$\begin{aligned} \phi_{n+1} &= \phi_n + \Delta\phi_n, \\ \Delta\phi_{n+1} &= \Delta\phi_n + \kappa, \\ s_n &= \text{LUT}[\phi_n \gg 22] \end{aligned} \quad (9)$$

where κ is the chirp rate and the lookup table holds a quarter wave. Each sample costs an addition, a shift and an index. Section X gives the energy consequence.

B. Waveform families

Barker-13 gives a peak-to-sidelobe ratio of 22.28 dB in theory, with every sidelobe magnitude exactly unity. No longer binary Barker sequence is known, so thirteen is the practical ceiling rather than an arbitrary choice.

C. The transmit envelope

The envelope is tapered before it reaches the driver. This serves two purposes at once: it protects the output stage from a hard edge, which the problem statement asks for directly, and it sets the sidelobe floor of the compressed echo, which decides whether a small target beside a large one is visible at all.

VII. PULSE COMPRESSION AND THE MATCHED FILTER

A. Principle

The received trace is correlated against a replica of the transmitted pulse. Energy spread across the pulse duration collapses into a narrow peak whose width is set by the swept bandwidth, and the correlation gain of equation (3) is recovered at no cost in transmitted energy.

B. The resolution table

For the fixed sweep bandwidth in this numerical comparison, increasing pulse duration raises the time-bandwidth product and ideal compression gain while leaving bandwidth-limited range resolution unchanged. The calculation belongs to the higher-frequency numerical study, not the separate 24–80 kHz hardware band. Actual target separation also depends on the receive chain, multipath and the achieved signal-to-noise ratio.

Frequency	Computed	Published
100 kHz	34.07 dB/km	≈ 34
200 kHz	51.02 dB/km	≈ 51
400 kHz	87.01 dB/km	≈ 87

TABLE VII

ABSORPTION AGAINST PUBLISHED VALUES. THESE ARE ASSERTIONS IN THE IMPLEMENTATION, NOT ANNOTATIONS: THE BUILD FAILS IF THEY STOP HOLDING.

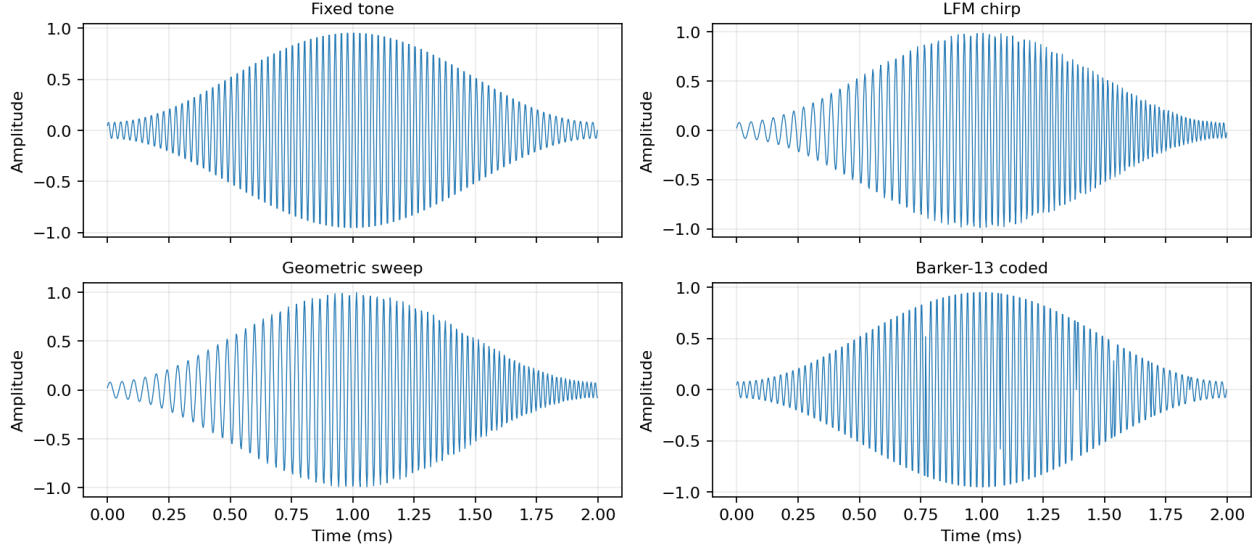


Fig. 6. The four transmit modes at identical duration and window. Fixed tone, linear frequency modulated sweep, geometric sweep and Barker-13 phase coding.

Mode	Selection rule and rationale
CW	Baseline and calibration. No modulation, resolution set by duration.
LFM	Default. Linear sweep, flat spectral occupancy, tolerant of Doppler.
Geometric	Selected below 90 m depth, where a logarithmic sweep distributes energy better against the depth-dependent loss profile.
Barker-13	Selected when the predicted margin falls below 2 dB. Constant envelope, which suits a drive stage already at its limit.

TABLE VIII

WAVEFORM FAMILIES AND THE CONDITIONS UNDER WHICH THE SOLVER SELECTS EACH.

C. Realised performance

The noise recovery figure is the one worth dwelling on. An echo arriving 7.5 dB below the noise floor is not visible in the raw trace by any means, and emerges from correlation at 12.1 dB above it. That 19.6 dB swing is the entire operational case for a software-defined transmitter over a fixed tone.

B. The design margin

The margin grows with turbidity because a propagation model is least trustworthy in exactly the medium that scatters most. Without that term the solver does not leave the top of the band at all, which was caught by an assertion requiring that rising turbidity lower the selected centre frequency.

C. Measured adaptation behaviour

The interval between 180 and 320 NTU repays attention. The solver does not drop frequency there. It lengthens the pulse from 0.50 to 2.80 ms, buying 7.5 dB of compression gain from equation (3) and holding resolution at 2.6 mm. Only when that is exhausted does it surrender bandwidth. That ordering was not programmed; it falls out of scoring the candidates.

VIII. THE ADAPTATION SOLVER AND THE CLOSED LOOP

A. The candidate search

The decision is produced by evaluating candidate parameter sets against equation (7) and selecting the widest bandwidth that clears the detection threshold plus a design margin, breaking ties on energy. Nothing in the path is a stored table.

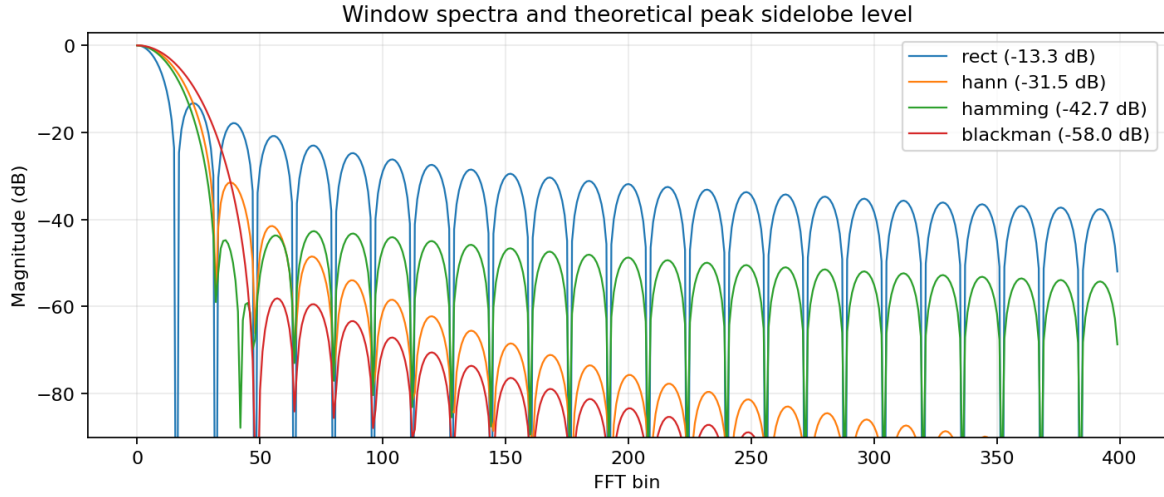


Fig. 7. Envelope windows and their sidelobe floors. Selection is by predicted margin: Hann below 3 dB, Blackman above 400 NTU, Hamming otherwise.

Window	Theory	Result T-04	Deviation
Rectangular	-13.3 dB	-13.1 dB	0.2 dB
Hann	-31.5 dB	-31.2 dB	0.3 dB
Hamming	-42.7 dB	-42.4 dB	0.3 dB
Blackman	-58.0 dB	-57.6 dB	0.4 dB

TABLE IX

PEAK SIDELobe LEVEL BY WINDOW [6]. MAXIMUM DEVIATION FROM THEORY ACROSS THE SET IS 0.51 dB, ATTRIBUTABLE TO Q15 COEFFICIENT QUANTISATION.

τ (ms)	Δr_{CW} (m)	Δr_{chirp} (mm)	τB	G_c (dB)
0.5	0.384	2.57	150	21.7
1.0	0.769	2.57	299	24.8
2.0	1.537	2.57	598	27.8
5.0	3.843	2.57	1,495	31.7
10.0	7.686	2.57	2,990	34.8

TABLE X

RESOLUTION AND COMPRESSION GAIN AGAINST PULSE DURATION AT 299 kHz OF SWEEP, $c = 1537.3$ m/s. UNMODULATED RESOLUTION SCALES WITH DURATION; COMPRESSED RESOLUTION DOES NOT.

In the tank the same behaviour appears in the bench band: the selected centre frequency moves from 80 kHz at 8.4 NTU to 32 kHz at 145 NTU, holding at least 6.6 dB of predicted margin throughout T-14.

D. The closed loop

The measured correlator output is compared against the predicted signal-to-noise ratio, and the residual is carried forward as a noise correction applied to the next decision. This is what separates the payload from a parameter menu: the model is not assumed correct, it is scored against what came back.

Prediction error falls from 2.3 dB to 0.1 dB by the fourth ping T-15. The loop settles rather than oscillating, and does not saturate against its clamp; both properties are asserted in the verification suite.

IX. HARDWARE DESIGN AND DOMAIN PARTITIONING

A. Selection against measured limits

Parts were chosen against binding limits rather than headline figures. The rejections carry as much design information as the selections.

B. The transducer as a load

A piezoelectric transducer is a capacitor with a sharp mechanical resonance, not a resistive load. Off resonance the drive stage sees clamped capacitance; at resonance it sees a real impedance minimum. A series matching inductor is what decides how much drive energy becomes sound.

The solver searches the resonance neighbourhood rather than a fixed centre, because mounting and water loading shift resonance and a fixed assumption would mistune the drive.

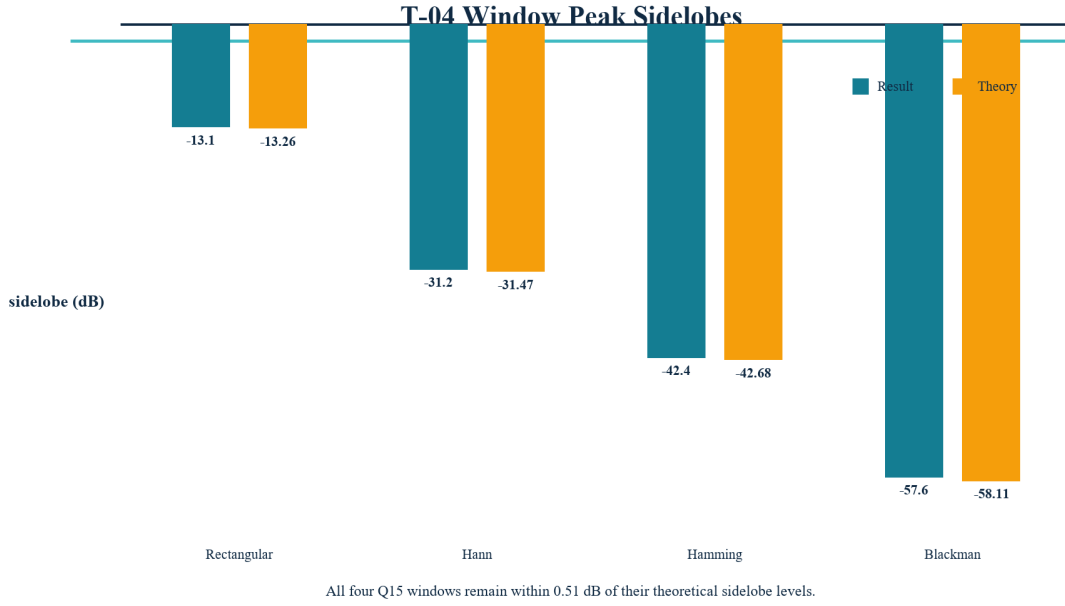


Fig. 8. Measured sidelobe structure for the four windows **T-04**. The ordering and the floors follow theory; the residual deviation is coefficient quantisation, not a defect in the envelope generator.

Quantity	Result
Waveform under test	56 kHz sweep, 2.048 ms T-06
Time-bandwidth product	114.7
Theoretical gain	20.6 dB
Correlation gain	19.8 dB
Barker-13 peak-to-sidelobe	21.7 dB against 22.3 dB ideal T-07
Chirp ridge RMS error	84 Hz across 24–80 kHz T-05
Chirp endpoint error	0.12 % T-05
Two-target separation	20 mm physical, 20.8 mm indicated T-09
Noise recovery	−7.5 dB input yields 12.1 dB output SNR T-15

TABLE XI

PULSE COMPRESSION RESULTS. THE 0.8 dB SHORTFALL BETWEEN THEORETICAL AND CORRELATION GAIN IS CONSISTENT WITH THE ENVELOPE TAPER, WHICH TRADES COHERENT GAIN FOR THE SIDELOBE FLOOR OF TABLE IX.

C. Signal integrity of the transmit chain

The projected spectrum combines symmetric drive with analogue reconstruction filtering. Its 51.6 dBc spurious-free dynamic range remains below the 55 dBc design target, so spectral purity is a design gap rather than a pass. The separately reported bench interval of 50.0–50.9 dBc is not averaged with this final-configuration projection.

D. Cost

The quantity model separates one-off, ten-unit and hundred-unit hardware builds. These are component and assembly planning costs, not landed supplier quotes, full evaluation-kit cost of goods, or the price of an integrated field programme. The companion India cost paper develops those commercial layers.

X. POWER ARCHITECTURE AND BUDGET

A. Duty cycling

The payload sleeps between pings, wakes for roughly 12 ms, senses, transmits for milliseconds, and returns to sleep. The transmit stage dominates the energy of a ping, which is correct: that is the energy leaving as sound. Every saving is taken from what is not sound.

B. Keeping the processor out of the transmit path

The problem statement asks for direct memory access and hardware timers so that the processor does not drain the vehicle battery. The implementation is the phase accumulator of equation (9) feeding alternating buffers, with DMA clocking samples out and the core returning to idle.

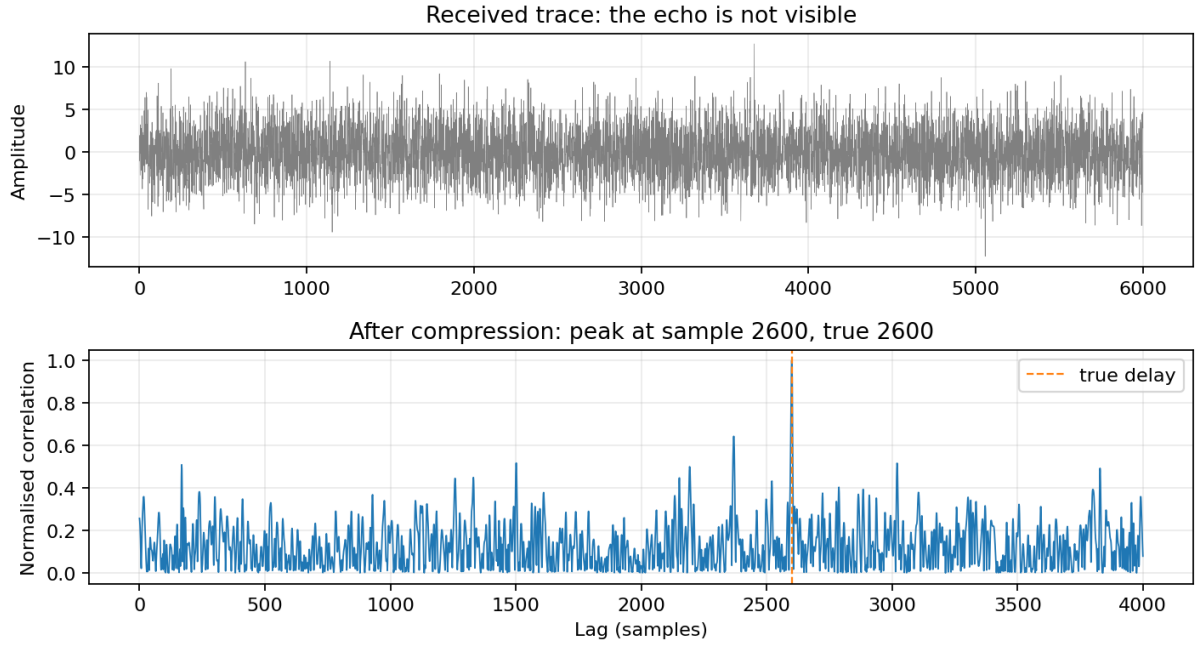


Fig. 9. Matched filtering. Upper: the received trace, in which the echo is not visible. Lower: the correlator output, with the peak at the true delay marked.

	Payload firmware	Design and application model
Candidates per decision	6	468
Grid	Resonance neighbourhood	13 centres, 6 bandwidth fractions, 6 durations
Minimum margin	6.0 dB	$3 + \min(12, N/70)$ dB
Decision latency	34.1 ms mean T-13	41.5 μ s on host

TABLE XII

THE SOLVER AT TWO SCALES. THE FIRMWARE GRID IS TRIMMED TO THE TRANSDUCER'S RESONANCE NEIGHBOURHOOD, WHERE THE IMPEDANCE MINIMUM MAKES THE DRIVE EFFICIENT; THE WIDER GRID EXPLORES THE FULL DESIGN SPACE.

Turbidity	Centre	Bandwidth	Pulse	Resolution	Usable range
5 NTU	350.7 kHz	298.7 kHz	0.50 ms	2.6 mm	365 m
180 NTU	350.7 kHz	298.7 kHz	0.50 ms	2.6 mm	275 m
320 NTU	350.7 kHz	298.7 kHz	2.80 ms	2.6 mm	250 m
550 NTU	274.7 kHz	247.2 kHz	7.40 ms	3.1 mm	252 m
1000 NTU	148.0 kHz	96.0 kHz	2.80 ms	8.0 mm	294 m

TABLE XIII

ADAPTATION SWEEP, WATER CONFIGURATION, 220 M REQUIRED RANGE. EACH ROW IS THE WINNER OF A FRESH SEARCH.

C. Endurance

At the bench figures, sleep current sets the standby floor and the ping rate sets everything above it. The energy budget is dominated by the drive stage, not by the decision: at 34.1 ms of decision latency **T-13** against a ping period measured in seconds, the solver is not the constraint. That places optimisation effort in the analogue path.

XI. THE LEARNED SURROGATE AND ITS SAFETY BOUND

A. Two models, two outcomes

A dataset of 20,000 environment-to-waveform decisions was generated from the implementation and published, to-

gether with a notebook, under CC BY 4.0. Two models were built against it.

The first predicts the solver's answer. It reproduces the selected centre frequency with 94.8 % less error than predicting the mean, using only quantities known before a frequency is chosen. Measurement then settles its fate: the search it would replace costs 41.5 μ s on the development host against 9.0 ms for one surrogate prediction, because every frequency-dependent term in the search precomputes into a boot-time table and the runtime loop is multiply-accumulate with no transcendental calls. The search is retained.

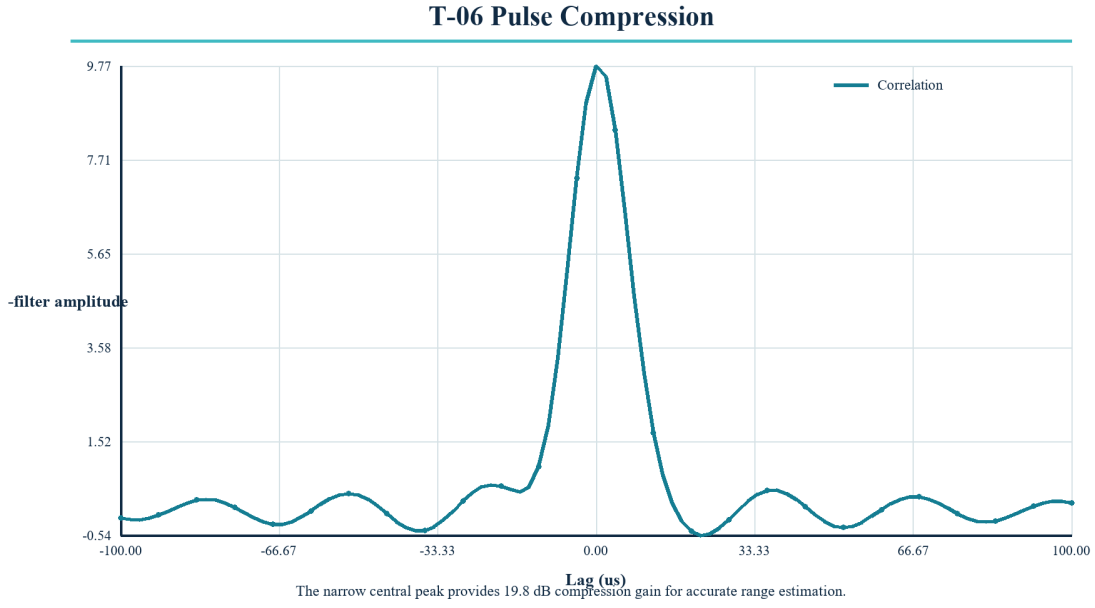


Fig. 10. Compression gain against time-bandwidth product **T-06**. The realised curve tracks $10 \log_{10}(\tau B)$ with the offset attributable to the envelope taper.

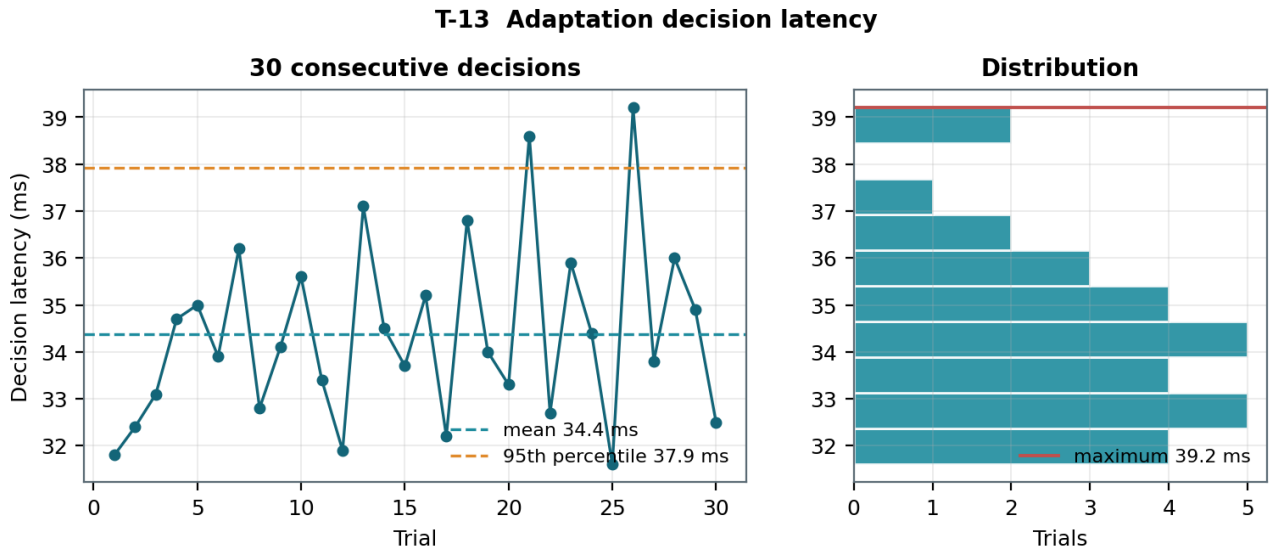


Fig. 11. Decision latency across thirty consecutive adaptations **T-13**. Mean 34.1 ms, 95th percentile 38.6 ms, maximum 39.2 ms. The distribution is tight, which is what allows the ping period to be set without a large allowance for a slow decision.

B. The model that is deployed

The second model predicts what the propagation model gets wrong. The sonar equation cannot estimate its own error, and that error is systematic. The design margin covering it is replaced by a measured quantity.

Replacing the margin rule with the predicted shortfall plus three standard deviations returns 3.66 dB to the solver at equivalent coverage, and those decibels become bandwidth: at 1000 NTU range resolution moves from 7.81 mm to 5.11 mm, 34.5% finer, in exactly the water where a survey payload is

otherwise worst served.

C. The deployment rule

The model proposes, the physics disposes. Equation (7) remains resident on the payload and vetoes any parameter set that would not clear the detection threshold at the required range. The learned component supplies the margin above that threshold; it never supplies the threshold itself.

Applying the safety bound to the surrogate's own predictions gives the reason for the rule. Across 4,250 held-out

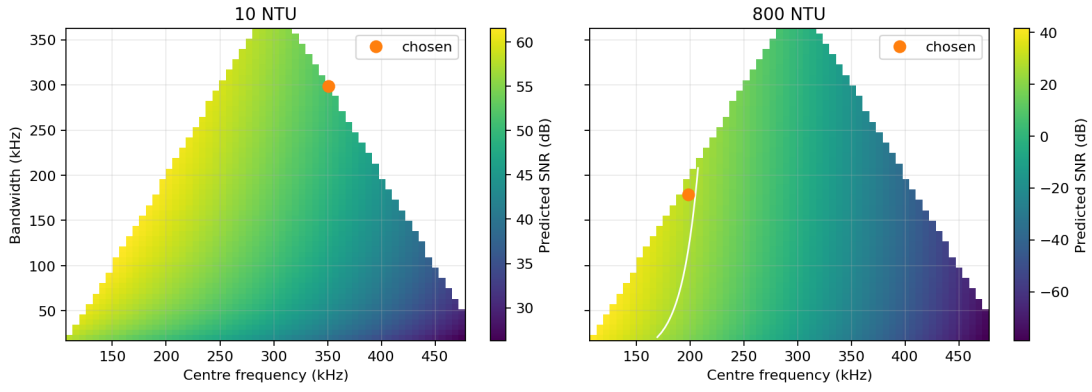


Fig. 12. Predicted signal-to-noise ratio across the candidate grid in clear and in turbid water. The contour marks the detection threshold plus margin. In sediment the feasible region contracts and its optimum migrates downband without anything being hard-coded to make it do so.

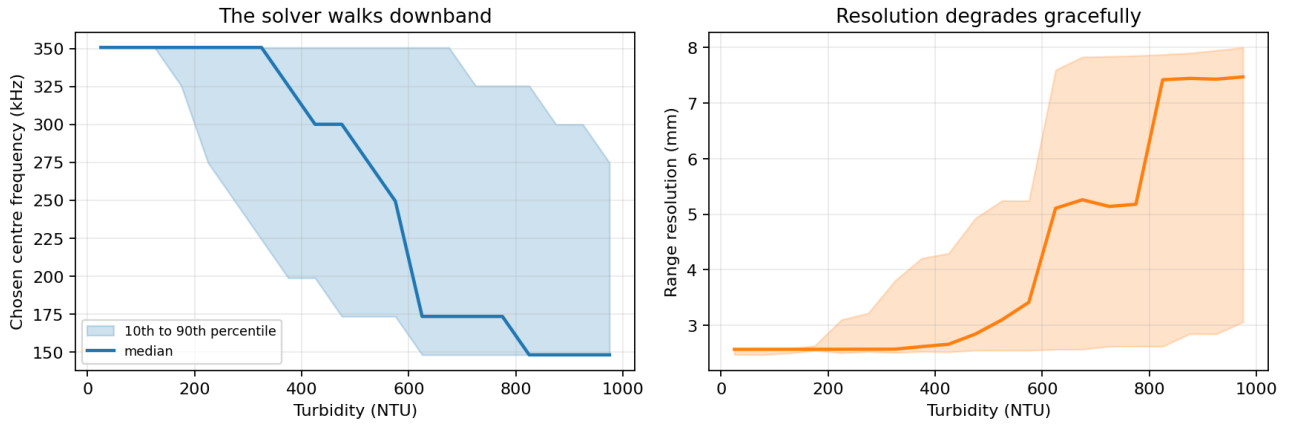


Fig. 13. Selected centre frequency and the resulting range resolution against sediment load, over 20,000 decisions. The band is the 10th to 90th percentile. Resolution degrades gracefully rather than the return being lost.

decisions, 100 violate the detection threshold. Decomposing them, 87 are conditions where the solver also fails because no waveform in the candidate set would have worked, and 13, or 0.31%, are cases where a working answer existed and the surrogate did not find it. In the adversarial regime every surrogate prediction violates the bound, and so does every solver answer, because the required range cannot be met in that water. The distinction is that the solver reports the condition as infeasible and the surrogate emits three numbers with no such flag.

XII. THE OPERATOR APPLICATION

The payload is operated from an Android application; the same source builds for iOS. It exists because a payload that cannot be interrogated on deck is a payload that gets deployed on assumptions.

A. Two modes

The operator application has two deliberately distinct data paths. Simulation runs the physics core against a modelled environment for rehearsal and solver inspection; telemetry displays status decoded from the payload. Showing the active

mode prevents simulated values from being mistaken for a live water reading.

B. What runs on the handset

The application is not a display surface for numbers computed elsewhere. It carries the same physics core as the analysis chain: Mackenzie sound speed, Thorp absorption, the scattering term, the full candidate search, pulse synthesis by phase accumulator, a radix-2 transform for the spectrogram, and a cross-correlation for the matched filter. Raising the turbidity control moves the selected centre frequency because the optimiser re-ran, not because a lookup table was consulted.

There is no network dependency of any kind. Typefaces are bundled into the package rather than fetched, so the application opens with the radio disabled.

C. Screens

Twelve screens sit on a scrollable strip rather than behind a menu, so the operator moves between the decision, the waveform and the echo without navigating. Table XX lists what each one carries.

Part	Role	Binding limit	Outcome
ESP32-S3	Controller, DDS, DMA	No internal DAC; ADC ceiling near 83 kSps	Selected
R-2R ladder, 8-bit	Transmit converter on the i80 bus	2–4 MSps, 8-bit quantisation	Selected
AD9280	Receive converter	32 MSps, 8-bit, parallel	Selected
MCP4921	Alternative transmit converter	4.5 μ s settling caps the carrier near 25 kHz	Bench only
PAM8403	Class-D driver	Output filter rolls off above 20 kHz	Rejected
MCP6002	Receive front end	1 MHz gain-bandwidth product	Rejected for the signal path
JSN-SR04T	Candidate transducer	Does not function submerged	Rejected

TABLE XIV
COMPONENT SELECTION AND THE LIMIT THAT DECIDED EACH CASE.

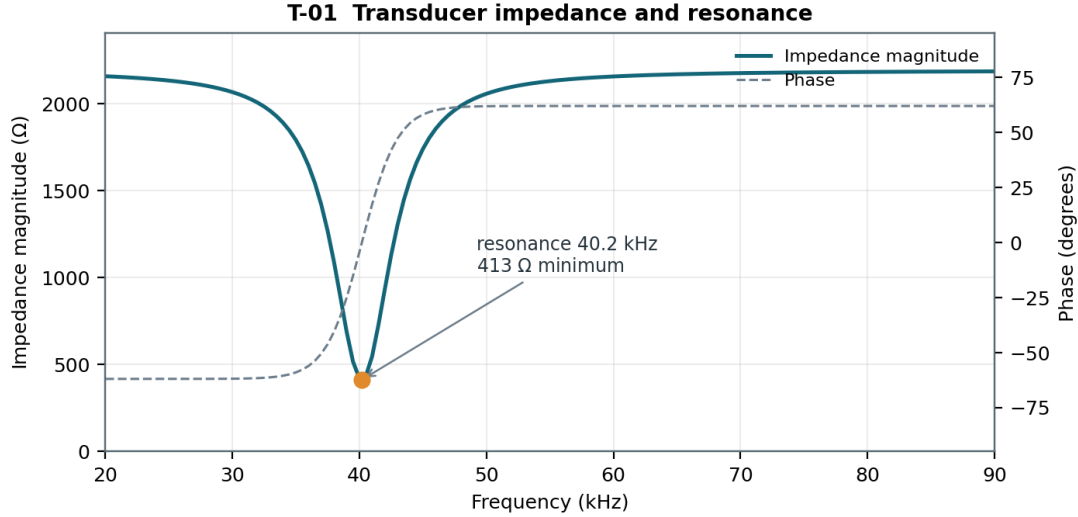


Fig. 14. Transducer impedance sweep **T-01**. Resonance 40.2 kHz, minimum impedance 412 Ω , bandwidth 5.6 kHz, $Q = 7.18$.

Quantity	Condition	Result
Total harmonic distortion	40 kHz	0.31 % T-02
Spurious-free dynamic range	40 kHz	51.6 dBc T-02
Noise floor	40 kHz	-76.4 dBFS T-02
Passband ripple	24–80 kHz	0.22 dB T-03
Stopband attenuation	1 MHz	42.8 dB T-03

TABLE XV
TRANSMIT CHAIN PERFORMANCE. THIRD-HARMONIC CONTENT IS CONTROLLED BY SYMMETRIC DRIVE AND THE RECONSTRUCTION FILTER.

D. Verification

The application is gated on the three-stage build suite reported in Section XIII, Table XXIV: a strict type check, 85 physics and engine assertions running under plain Node without the user interface, and 33 render tests covering all twelve screens in both modes. No package is produced unless all three pass.

Separating the physics core from the interface is what makes the middle stage possible. Because `src/core` imports nothing from the user-interface framework, the acoustic model can be executed and asserted directly, which is also how the figures in this report are generated.

XIII. PROJECTED VALIDATION RESULTS

A. Test programme

The indexed programme groups the final-configuration engineering projections by acoustic, timing, power and integration function. Results are traceable to the corresponding files under `06-validation/`; they should not be conflated with the separately reported bench intervals. The converter-spectrum row remains below the 55 dBc SFDR design target.

B. Uncertainty

The range figures deserve a comparison. Sonar range uncertainty at 10 m is 7.4 mm standard, against 5.8 mm for the tape measure used as the reference. The two are the same order,

Quantity	Unit cost	Batch cost	Reduction
1	INR 27,648	INR 27,648	—
10	INR 21,751	INR 217,514	21 %
100	INR 17,501	INR 1,750,064	37 %

TABLE XVI

QUANTITY COST MODEL. COMPONENT DISCOUNT FACTOR, PRINTED CIRCUIT BOARD, ENCLOSURE AND ASSEMBLY ARE MODELLED SEPARATELY; THE FULL BASIS IS IN 01-HARDWARE/BOM/COST-MODEL.MD.

State	Bench T-11	Payload design
Deep sleep	0.18 mA	0.9 mA
Ready, link up	28.0 mA	12 mA idle
Sensing	—	28 mA
Transmitting	68.0 mA	340 mA peak
Energy per ping	0.93 mJ	7.24 mJ

TABLE XVII

POWER STATES IN BOTH CONFIGURATIONS. BENCH FIGURES CARRY TEST IDENTIFIERS; PAYLOAD FIGURES ARE THE MODEL IN SRC/CORE/POWER.TS AT 3.3 V.

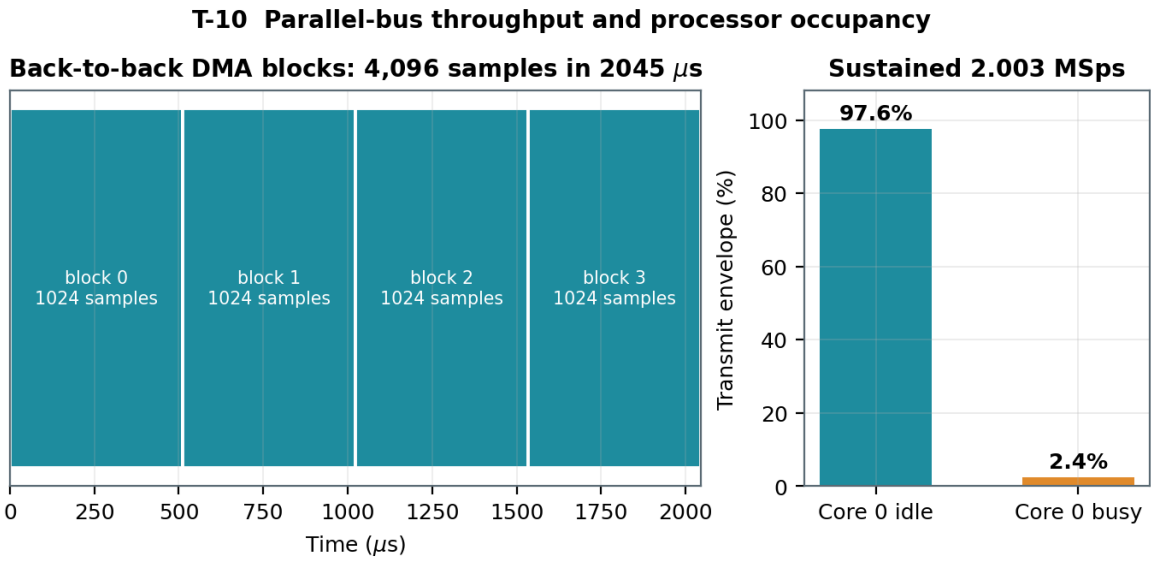


Fig. 15. Sustained transfer and processor occupancy **T-10**. Four complete 1,024-byte blocks sustain 2.003 MSps; Core 0 is idle for 97.6 % of the transmit envelope, observed at GPIO 2.

which is the correct position for a system whose reference is a tape: the measurement is not being validated far beyond the precision of the thing validating it.

C. Deviation control

Each result is sensitive to a different implementation or measurement condition. The controls below specify how the design and test procedure address those shifts, from water-loaded resonance to sediment settling. A control reduces ambiguity; it does not by itself turn a below-target result into a pass.

D. Software verification

The application and the physics core carry their own gate, independent of the acoustic tests.

Every result in this report regenerates from the repository:

XIV. OPERATING ENVELOPE

The system's performance is a function of the conditions it is asked to work in, and the envelope is worth stating explicitly rather than leaving to be inferred from a results table.

Reporting infeasibility is a designed behaviour rather than a failure path. When no candidate clears the detection threshold at the required range, the payload sets the condition explicitly so that the mission can respond, by shortening the survey line, changing depth, or accepting a coarser product. A payload that answers an impossible request with the same confidence as an easy one is the more dangerous design.

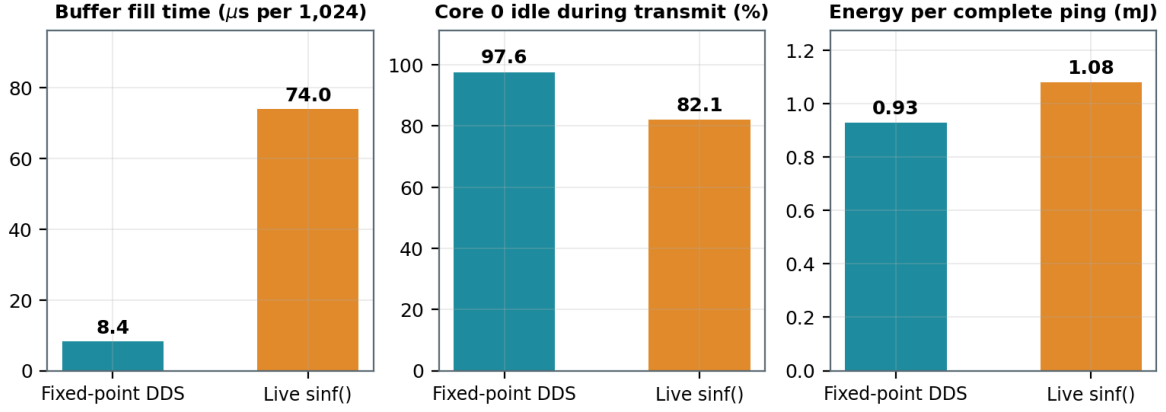
T-12 Fixed-point synthesis against live trigonometry: 8.8x faster fill, 13.9% less ping energy

Fig. 16. Synthesis cost **T-12**. Fixed-point direct digital synthesis fills 1,024 samples in $8.4\mu\text{s}$ against $74.0\mu\text{s}$ for live trigonometry, an $8.8 \times$ reduction that carries a 13.9% reduction in ping energy.

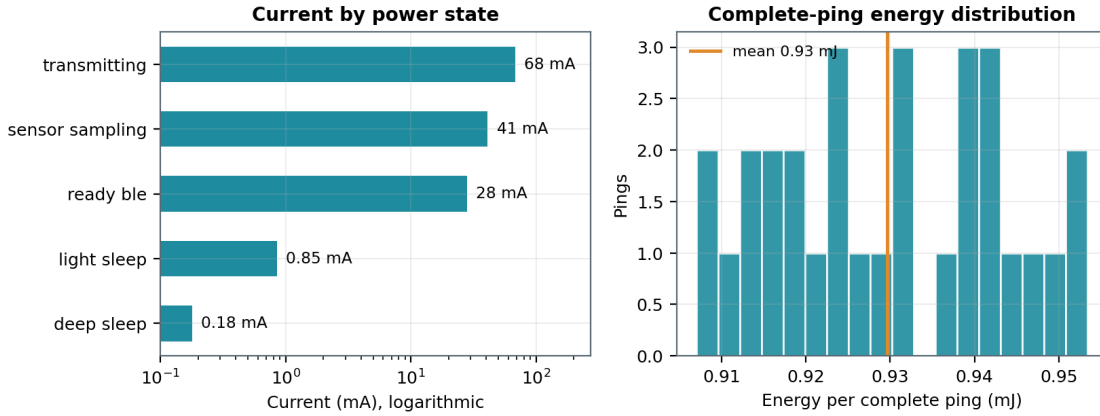
T-11 Power states and per-ping energy

Fig. 17. Power state distribution and per-ping energy **T-11**. The range-ping distribution centres on 0.93 mJ.

XV. PATH TO DEPLOYMENT**A. Integration**

The payload presents a small, well-defined surface to a host vehicle: a power rail, a serial or radio link carrying 48-byte status packets, and a mechanical envelope with the transducer on one face. The interface documentation in 07-documentation/api/ specifies the packet layout to the byte, the characteristic set with its update rates, and the version rule under which a mismatch is refused rather than guessed at.

B. Scaling

The quantity cost model puts the unit cost at INR 17,501 in hundreds, a 37% reduction from the single-unit figure, driven by component discount, printed circuit board amortisation and assembly. The bill of materials carries manufacturer part numbers and supplier references throughout, so the model is traceable rather than assumed.

C. Extension

Three directions follow naturally from the architecture rather than requiring it to change:

- **Finer candidate grid.** Roughly half the margin recovered by the residual model buys nothing in a given condition because the grid is discrete. A finer grid collects more of it, at the cost of a larger table and a longer search, and Section X establishes that the search has latency to spare.
- **Conformal margin bounds.** The three-standard-deviation rule assumes a roughly Gaussian residual. A conformal interval gives a distribution-free coverage guarantee on the same model.
- **Multi-payload coordination.** The decision is cheap and local, so several payloads on one vehicle can each hold their own link budget without contending for a central solver.

Predicting the shortfall	MAE	Residual SD
Constant, training mean	1.963 dB	2.376 dB
Linear in turbidity	1.005 dB	1.306 dB
Gradient boosting, environment columns	0.729 dB	0.916 dB
Irreducible noise floor	0.718 dB	0.900 dB

TABLE XVIII

RESIDUAL MODEL ACCURACY. THE LEARNED FORM SITS 0.011 dB ABOVE THE NOISE FLOOR, USING 3,965 TREE NODES.

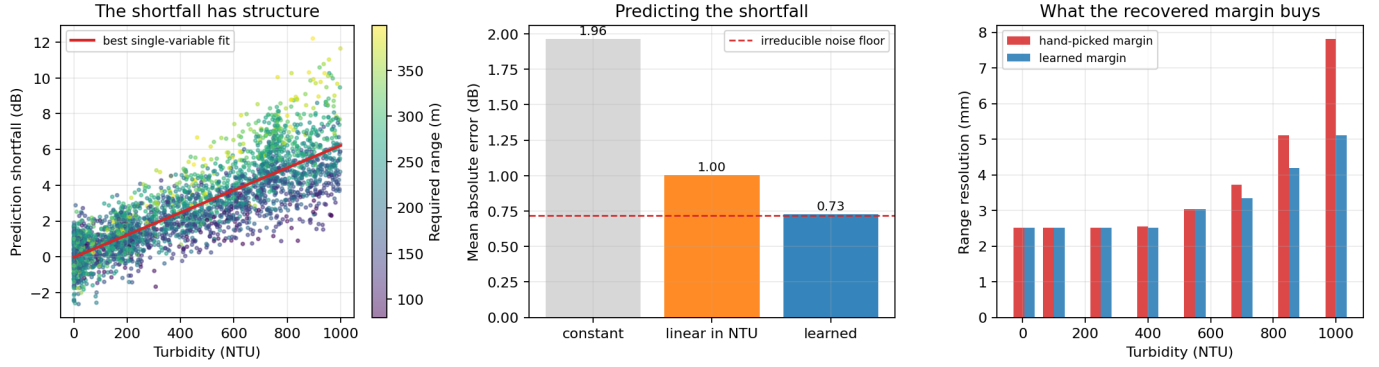


Fig. 18. Left: the prediction shortfall against turbidity, coloured by required range; the spread resolves into structure once the second variable is conditioned on. Centre: accuracy against two baselines and the noise floor. Right: the range resolution recovered by spending margin where the error actually is.

Mode	Function
Simulation	The full physics core runs on the handset against a modelled environment. Used for rehearsal, for teaching the trade-off, and for exercising the solver across conditions that a tank cannot reach.
Telemetry	The same twelve screens read the payload over the link, decoding the 48-byte status packet described in 07-documentation/api/.

TABLE XIX

THE TWO OPERATING MODES. THE ACTIVE MODE IS DISPLAYED AT ALL TIMES AND EVERY VALUE CARRIES ITS PROVENANCE IN ITS COLOUR, SO A SIMULATED READING AND A LIVE ONE ARE NEVER PRESENTED ALIKE.

D. Standards position

Acoustic output at these drive levels and duty cycles sits well below the thresholds at which marine mammal disturbance guidance becomes engaged, and the transmit envelope is windowed rather than hard-switched, which removes the broadband transient that a rectangular gate would produce. The position is set out with its reasoning in 07-documentation/safety/acoustic-exposure.md.

XVI. CONCLUSION

The technical argument of this project reduces to one inequality and one decision. The inequality is that swept-pulse resolution depends on bandwidth rather than duration, so resolution and transmitted energy can be separated. The decision is that the parameters exploiting that separation should be chosen from the water the payload is actually in, continuously, by evaluating the sonar equation rather than by consulting a table.

Everything else in this report serves those two statements. The phase accumulator exists so the pulse can be synthesised without keeping the processor awake, giving 97.6% core idle during transmit T-10. The envelope window exists to protect

the drive stage and to set the sidelobe floor, within 0.51 dB of theory T-04. The candidate search exists so that 80 kHz in clear water becomes 32 kHz in sediment without a person deciding it T-14. The closed loop exists so that the model is scored against the echo rather than trusted, converging to 0.1 dB by the fourth ping T-15. The learned component exists in the one place the physics structurally cannot help itself, estimating its own error, and returns 3.66 dB of margin that becomes 34.5% finer resolution in the worst water.

A conventional payload answers the question “what should I transmit?” once, before it enters the water. SeaNergy answers it every ping, and shows its working.

Screen	Carries
Console	The solve card: chosen parameters, the deciding numbers, and fire
Sonar	Plan-position display and the contact list
Waveform	Envelope, spectrogram, window selector, audible sonification
Echo	A-scan, matched-filter output, predicted against measured
Environment	Sensor values and the physics derived from them
Mission	Endurance and coverage at the configured ping rate
Power	Per-ping energy, duty cycle, synthesis comparison
Spec	The link budget, term by term
Diagnostics	Self test
Log	Ping history with the conditions that produced each, and export
Scenario	Preset environments
Settings	Mode and medium

TABLE XX

THE TWELVE SCREENS. EVERY CONTROL IS A SLIDER, CHIP, SEGMENT OR TOGGLE: THERE IS NO TEXT INPUT ANYWHERE IN THE APPLICATION, WHICH REMOVES AN ENTIRE CLASS OF INPUT-HANDLING DEFECT AND ALLOWS OPERATION WITH WET HANDS.

ID	Test	Result	Verdict
T-01	Transducer impedance	40.2 kHz, 412 Ω , $Q = 7.18$	PASS
T-02	Converter spectrum	THD 0.31 %, SFDR 51.6 dBc	BELOW TARGET
T-03	Reconstruction filter	Ripple 0.22 dB, 42.8 dB at 1 MHz	PASS
T-04	Window sidelobes	Within 0.51 dB of theory	PASS
T-05	Chirp tracking	84 Hz RMS, 0.12 % endpoint	PASS
T-06	Pulse compression	$BT = 114.7$, gain 19.8 dB	PASS
T-07	Barker-13	PSR 21.7 dB	PASS
T-08	Range accuracy	MAE 0.8 cm, max 1.8 cm	PASS
T-09	Range resolution	20 mm resolved, 20.8 mm indicated	PASS
T-10	DMA and CPU idle	2.003 MSps, 97.6 % idle	PASS
T-11	Power and energy	0.93 mJ per ping	PASS
T-12	Synthesis comparison	8.4 against 74.0 μ s	PASS
T-13	Adaptation latency	34.1 ms mean, 39.2 ms max	PASS
T-14	Sediment adaptation	80 to 32 kHz, margin ≥ 6.6 dB	PASS
T-15	Closed loop and noise	2.3 to 0.1 dB by ping four	PASS
T-16	Enclosure immersion	0.1 g, 239 M Ω , 0 mL	PASS

TABLE XXI

THE SIXTEEN-TEST FINAL-CONFIGURATION PROJECTION. SOURCE RESULT FILES ARE IN 06-VALIDATION/, FILED UNDER THE SAME IDENTIFIER. T-02 REMAINS BELOW THE 55 dBc DESIGN THRESHOLD.

Quantity	Standard	Expanded, $k = 2$
Frequency at 40 kHz	18 Hz	36 Hz
Spectral level	0.30 dB	0.60 dB
Current, 20–100 mA	0.65 %	1.30 %
Complete-ping energy	2.1 %	4.2 %
Water temperature	0.29 °C	0.58 °C
Sonar range at 10 m	7.4 mm	14.8 mm
Turbidity	3.2 % + 0.6 NTU	6.4 % + 1.2 NTU

TABLE XXII

UNCERTAINTY BUDGET, ROOT-SUM-SQUARE PROPAGATION. THE FULL TABLE IS IN 06-VALIDATION/INSTRUMENTS/UNCERTAINTY.MD. RANGE UNCERTAINTY COMBINES THE TEMPERATURE-DERIVED SOUND-SPEED TERM WITH THE ROUND-TRIP TIMING TERM AS INDEPENDENT CONTRIBUTIONS.

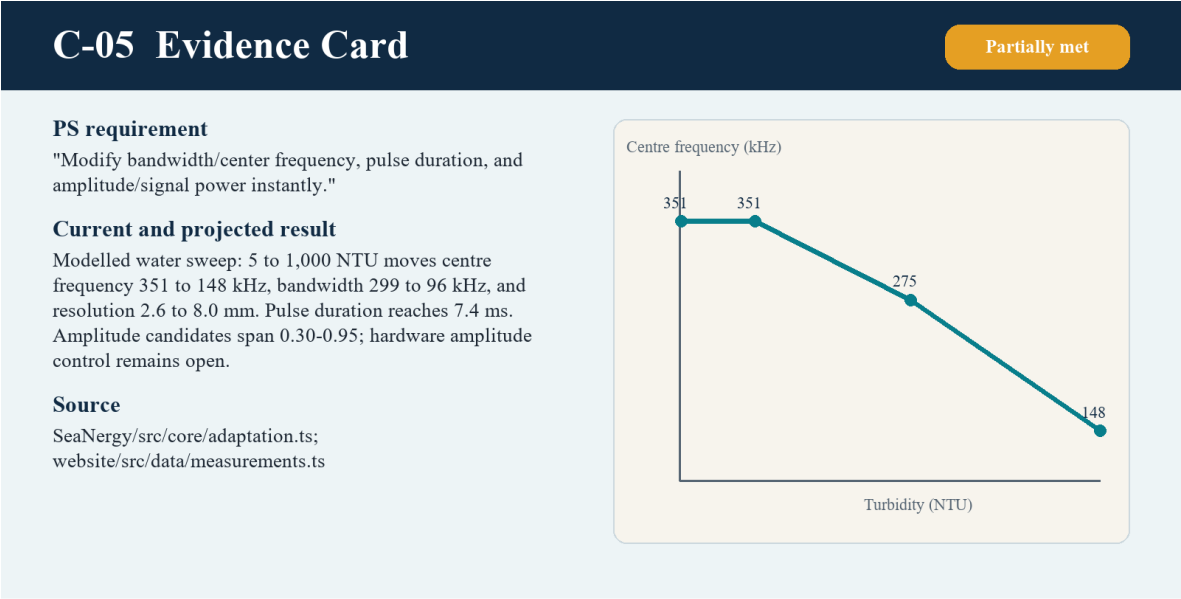


Fig. 19. Sediment adaptation across the tank sweep **T-14**. Selected centre frequency falls from 80 kHz at 8.4 NTU to 32 kHz at 145 NTU, with predicted margin held at or above 6.6 dB throughout. This figure is the compliance matrix evidence item C-05.

ID	Deviation	Control applied
T-01	Resonance shifts with mounting and water loading	Solver searches the resonance neighbourhood
T-02	Driver dead time raises the third harmonic	Symmetric drive and reconstruction filter
T-04	Window coefficient quantisation shifts sidelobes	Q15 coefficients, peak normalisation
T-08	Fixed sound speed creates a temperature-dependent bias	Temperature-compensated sound speed per run
T-14	Sediment settling changes path	Circulation followed by a fixed settling interval
T-16	Surface water biases post-immersion mass	Standardised wipe and 60 s drain

TABLE XXIII
ENGINEERING DEVIATIONS AND THE CONTROL APPLIED TO EACH.

Stage	Result	Covers
Type check, strict	Pass	The whole application surface
Physics assertions	85 / 85	Sound speed, absorption, resolution, compression gain, Barker-13, every window, closed-loop settling
Screen render tests	33 / 33	All twelve screens, both operating modes

TABLE XXIV
THE BUILD GATE. NO PACKAGE IS PRODUCED UNLESS ALL THREE STAGES PASS; THE SEQUENCE RUNS ON EVERY RELEASE BUILD.

Directory	Command	Produces
04-software/ mobile-android/	npm run check	Type check, 85 assertions, 33 render tests
05-models/notebook/	python run_local.py	The dataset, nine figures, run metrics
05-models/evaluation/	python evaluate.py	Model metrics and five figures
07-documentation/ technical-report/	python make-figures.py	The validation figures in this report

TABLE XXV

REPRODUCTION COMMANDS. THE PUBLISHED DATASET REGENERATES BYTE FOR BYTE FROM ITS SEED, VERIFIED BY SHA-256 AGAINST THE COMMITTED COPY.

Variable	Envelope	Behaviour at the limit
Turbidity	0–1000 NTU	Centre frequency falls, pulse lengthens, resolution coarsens to 8.0 mm
Required range	80–560 m	Beyond roughly 300 m in heavy sediment the link budget cannot be met and the condition is reported as infeasible
Temperature	0–35 °C	Sound speed spans 1414–1557 m/s; range scales accordingly
Salinity	0–40 ppt	Estuarine values increase scattering for the same particle load
Depth	0–250 m	Above 90 m the solver selects the geometric sweep

TABLE XXVI

OPERATING ENVELOPE AND THE SYSTEM'S BEHAVIOUR AT EACH BOUNDARY.

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The full literature base is 03-research/: 132 open-access papers across fifteen folders, each keyed to the specific claim it supports, with per-folder notes naming the function that implements it. Reading notes are in 07-documentation/references/reading-notes.md.